Denavit-Hartenberg Approach to Forward Kinematics

Summary:

This lesson introduces the Denavit-Hartenberg (DH) method, a repeatable recipe for forward kinematics in which every joint becomes one matrix. On Day 1, students write rotation and movement as matrices and chain them to find the end-effector position of the simplified 2-link planar arm from Lesson 3. On Day 2, students extend the recipe to the 3D arm: they place a coordinate frame on each joint, fill in a DH table, build one matrix per joint, and multiply them. By the end of the lesson, students can calculate the predicted end-effector position for a given set of joint angles and confirm that it matches the trigonometric result from Lesson 3.

Objective:

By the end of this lesson, students will be able to:

- Write a rotation and a movement as matrices and chain them to find the end-effector position of the 2-link planar arm
- Explain why the order of matrix multiplication matters
- Place a coordinate frame on each joint and fill in the DH table (θ, d, a, α) for the base, shoulder and elbow
- Build one DH matrix per joint and multiply them to calculate the 3D end-effector position
- Verify the DH result against the Lesson 3 trigonometric result and compare the two methods

Standards:

NGSS

- Engineering Design: HS-ETS1-2 (break a complex problem into smaller problems), HS-ETS1-3 (evaluate solutions against trade-offs), HS-ETS1-4 (use a computer model to test a solution)
- Science & Engineering Practices: Developing and Using Models; Using Mathematics and Computational Thinking; Engaging in Argument from Evidence
- Crosscutting Concepts: Systems and System Models; Patterns

Vocabulary:

Forward kinematics: Finding the end effector's position from the joint angles.

End effector: The tip of the arm. It is the point whose position we calculate.

Rotation matrix: A matrix, R(θ), that spins a point by θ around the origin, counter-clockwise.

Move matrix: A matrix, T(dx, dy), that slides a point by dx and dy. It works only because the point is written with an extra 1.

Extra 1: Writing the point (x, y) as (x, y, 1), or (x, y, z) as (x, y, z, 1), so that moving a point becomes a matrix multiplication.

Chain of matrices: One matrix for each step of the arm, multiplied in the order of the joints to give a single matrix for the whole arm.

Coordinate frame: A set of perpendicular x, y and z axes attached to a joint that moves with it.

Denavit-Hartenberg (DH) method: A standard recipe for placing a frame on each joint and describing the step from one frame to the next with four numbers.

DH numbers (θ, d, a, α): The four numbers that describe one joint: θ is the turn about z, d is the slide along z, a is the slide along x (the link length) and α is the turn about x (the link twist).

DH table: A table with one row per joint, listing its θ, d, a and α.

Transformation matrix: A 4 × 4 matrix that combines a turn and a move. Its last column holds the position of the new frame's origin.

Resources & Materials:

- Lesson 4 slideshow, Days 1 and 2 (37 slides; the slide numbers in the Procedures refer to it)
- Calculators for the Try It Yourself problems
- The assembled 4-DOF arm from Lesson 2, to point out each joint's turning axis (optional)
- Computers with Python and NumPy, plus the helper file `dh_helper.py (code in the Appendix)`, for the "Check It with Code" slide.

Procedures:

Day 1: 2D Forward Kinematics with Matrices

- **Opening (4 min)** — Display the title slide, then "Where Lesson 3 Left Off" and "One Recipe for Every Joint" (slides 1–3). Recall that Lesson 3's triangles gave (X, Y) = (426.78, 176.78) for θS = 45° and θE = 45°, but every new joint needed new triangles. Today's goal is a recipe where each joint becomes one matrix, so a computer can do the work. Note that θS (shoulder), θE (elbow) and θB (base) replace Lesson 3's θ1, θ2 and θ3.
- **Rotating a point (4 min)** — Show "Rotating a Point" (slide 4): spin P = (250, 0) by 45° using x′ = 250cos45° and y′ = 250sin45° to get P′ = (176.78, 176.78). Point out that these are the same numbers as ΔX1 and ΔY1 in Lesson 3 — the first link is just a point that has been rotated.
- **The rotation matrix (8 min)** — Show that the two formulas fit inside one matrix R(θ), and explain how to multiply a row of the matrix by the point ("The Rotation Matrix", slide 5). Walk through R(45°) · (250, 0) = (176.78, 176.78) and confirm it matches the cos/sin result ("Worked Example: Rotation", slide 6).
- **Try It Yourself #1 (5 min)** — Students rotate P = (250, 0) by 90° and then by 30° (slide 7); reveal solutions (0, 250) and (216.51, 125.00).
- **Moving a point and the extra 1 (8 min)** — Show that rotating is multiplying but moving is adding, so a 2×2 matrix cannot move a point ("Moving a Point", slide 8). Introduce the extra 1: writing (x, y) as (x, y, 1) lets a 3×3 matrix T(dx, dy) move a point, and rotation becomes a 3×3 matrix too ("The Extra 1", slide 9). Demonstrate T(100, 0) · (250, 0, 1) = (350, 0, 1).

- **The arm as a chain of matrices (7 min)** — Describe the arm as a story told from the base outward: turn at the shoulder, travel along the upper arm, turn at the elbow, travel along the forearm. As matrices, M = R(θa) · T(250, 0) · R(θb) · T(250, 0) ("The Arm as a Chain of Matrices", slide 10). Convert Lesson 3's angles, since matrices count counter-clockwise from the x-axis: θa = 90° − θS and θb = −θE ("Converting the Arm's Angles", slide 11).
- **Worked example (9 min)** — Walk through θS = 45° and θE = 45° ("Worked Example: Setup" and "Worked Example: Right to Left", slides 12–13): convert the angles to θa = 45° and θb = −45°, then multiply from the right, starting at the point (0, 0, 1) and applying T(250, 0), R(−45°), T(250, 0) and R(45°) in turn. Finish with the "Completed Variables" slide (slide 14): (X, Y) = (426.78, 176.78), matching Lesson 3's (426.777, 176.777). The story of the arm reads left to right, but the arithmetic starts at the right, next to the point (see the note on slide 12).
- **Try It Yourself #2 and #3 (10 min)** — Students solve θS = 30°, θE = 30° and reveal solution (341.51, 341.51); then solve θS = 60°, θE = 0° and reveal solution (433.01, 250.00) (slides 15–16). These are Lesson 3's two Try It Yourself problems, so students can compare against the triangle answers. Use #3 to show that a 0° turn changes nothing.
- **Order matters (3 min)** — Show "Order Matters" (slide 17): rotating and then moving ends at (100, 250), but moving and then rotating ends at (0, 350). The same two matrices in a different order give a different answer, so the matrices in the arm's chain must follow the order of the joints.
- **Recap & wrap-up (2 min)** — Show the "Day 1 Recap" slide (slide 18) as a summary of rotate, move, chain and order, then preview Day 2: the real arm's base turns too, so we need 3D.

Day 2: Extending to 3D with the DH Method

- **Recap (3 min)** — Show "Extending to 3D" (slide 19) and review the Day 1 chain, M = R(θa) · T(250, 0) · R(θb) · T(250, 0), which gave (426.78, 176.78). Quick-fire questions: Does the order of the matrices matter? Yes. What is θa for θS = 30°? (60°.)
- **The base turns the whole arm (4 min)** — Present "The Base Turns the Whole Arm" (slide 20): the base spins Day 1's flat arm around the vertical axis by θB, so a point is now (x, y, z) and each matrix is 4×4. Note the change of habit: from now on z points up, whereas Lesson 3 called the up-direction Y.
- **The four DH numbers (7 min)** — Introduce the four numbers that describe each joint: θ (turn about z), d (slide along z), a (slide along x) and α (turn about x), with the values for the base, shoulder and elbow ("Four Numbers Describe Each Joint", slide 21). Show the matrix A = Rz(θ) · Tz(d) · Tx(a) · Rx(α) ("One Matrix per Joint", slide 22): the top-left 3×3 shows how the new frame is turned and the last column shows where its origin is. Emphasize that students never expand it by hand; they plug in one row of the DH table.
- **Placing frames on the arm (8 min)** — Teach the two rules ("Placing a Frame on Each Joint", slide 23): z points along the axis the joint turns around, and x points along the link, away from the previous joint. In pairs, students sketch the frames on a side view of the arm (base, shoulder, elbow and tip), then reveal "Frames on Our Arm" (slide 24). Common mistake: drawing z along the links instead of along the joint axes. Teacher note: "x points along the link" is a simplification

that works for this arm. The full DH rule is that x is perpendicular to both z axes. This lesson uses the standard DH form, A = Rz(θ) · Tz(d) · Tx(a) · Rx(α); some textbooks use the modified (Craig) form, whose table and matrix look different.

- **Filling in the DH table (7 min)** — Pairs fill in the blank table row by row ("Fill In the DH Table", slide 25), asking the four questions on the slide for each joint; reveal "The DH Table for Our Arm" (slide 26). Explain the θ column: θ = 90° − θS for the shoulder and θ = −θE for the elbow, the same conversions as Day 1, and α = 90° appears only in the base row.
- **Building the matrices (5 min)** — Put the pose θB = θS = θE = 45° into each row of the table to build A_1, A_2 and A_3 ("A_1: The Base", "A_2: The Shoulder" and "A_3: The Elbow", slides 27–29). Read each last column: (0, 0, 0) for A_1, (176.78, 176.78, 0) for A_2 and (176.78, −176.78, 0) for A_3.
- **Multiplying the matrices (9 min)** — Multiply $A_2 \cdot A_3$ first ("Step 1", slide 30): the turns cancel, and the last column (426.78, 176.78, 0) is Day 1's answer. Then multiply $A_1 \cdot (A_2 \cdot A_3)$ ("Step 2", slide 31): the last column of T is the tip position (x, y, z) = (301.78, 301.78, 176.78). Close with "Same Answer as Lesson 3" (slide 32): Lesson 3 put up on Y, so swap its Y and Z.
- **Try It Yourself #4 (5 min)** — Students fill in the θ column and find the tip position for θB = 30°, θS = 60°, θE = 0° (slide 33); reveal solution (375.00, 216.51, 250.00). It is the same arm shape as Try It Yourself #3 with the base turned 30°.
- **Try It Yourself #5 (4 min)** — Students solve θB = 120°, θS = 0°, θE = 90° (slide 34); reveal solution (−125.00, 216.51, 250.00). Discuss why x is negative (the base has turned past the y axis) and why z = 250 (the upper arm stands straight up and the forearm lies flat). If time is short, assign this problem as homework.
- **Check it with code (3 min)** — Run the code on "Check It with Code" (slide 35), which prints `[301.78, 301.78, 176.78]` for the worked example. Students change the three angles to the values from Try It Yourself #4 and #5 to check their answers. The helper file does the matrix arithmetic; students write this code themselves in Lesson 5.
- **Trigonometry vs. DH (3 min)** — Show "Trigonometry vs. DH Matrices" (slide 36). Ask: if the arm had six joints, which method would you use, and why? Both methods give the same answer, but DH keeps working as the arm gets bigger. If the lesson runs long, this step can be shortened to a one-minute summary.
- **Wrap-up (2 min)** — Show "What We Learned" (slide 37) as a summary: every joint becomes one matrix built from θ, d, a and α; z is the joint axis and x is the link; multiply the matrices in joint order; the last column of T is the tip position. Preview Lesson 5, where forward kinematics becomes code.

Assessment:

Complete the Try It Yourself problems in the slideshow (#1–#5). Students check each answer against the solution on the slide and can verify #4 and #5 with the code on the "Check It with Code" slide.

Use these questions as an exit ticket or short quiz. The answers are in red.

1. Rotate the point P = (100, 0) by 90°. What is P′?
   Answer: P′ = (0, 100)

2. Why can a 2 × 2 matrix not move a point, and what trick fixes this?
   Answer: Moving is adding, and a matrix can only multiply. Writing the point with an extra 1, as (x, y, 1), lets a 3 × 3 matrix do the move.

3. True or False: R(90°) · T(100, 0) gives the same result as T(100, 0) · R(90°) for the point P = (250, 0).
   Answer: False. T(100, 0) · R(90°) · P rotates first and ends at (100, 250). R(90°) · T(100, 0) · P moves first and ends at (0, 350).

4. Convert θS = 30° and θE = 45° to the matrix angles θa and θb.
   Answer: θa = 90° − 30° = 60° and θb = −45°

5. Name the four DH numbers and say what each one does.
   Answer: θ turns about z, d slides along z, a slides along x (the link length) and α turns about x (the link twist)

6. Give the DH table for our arm (base, shoulder, elbow): the θ, d, a and α for each row.
   Answer: Base: θ = θB, d = 0, a = 0, α = 90°. Shoulder: θ = 90° − θS, d = 0, a = 250, α = 0°. Elbow: θ = −θE, d = 0, a = 250, α = 0°

7. Find the tip position (x, y, z) for θB = 90°, θS = 90° and θE = 0°.
   Answer: (0, 500, 0). The θ column is 90°, 0°, 0°, so the upper arm lies flat, the straight forearm adds another 250 mm, and the base turn of 90° points the whole arm along the y axis

8. Which part of $T = A_1 \cdot A_2 \cdot A_3$ gives the tip position?
   Answer: The top three entries of the last column

9. In one or two sentences, why would an engineer choose the DH method over the triangles of Lesson 3?
   Answer: DH uses the same recipe for every joint, so it keeps working as the arm gets bigger, and a computer can do the multiplying. Triangles need new drawings for every new joint

Appendix: dh_helper.py

Save this code as dh_helper.py in the same folder as the student code. The “Check It with Code” slide imports dh_matrix and position from it.

```
"""Helper functions for Lesson 4: the Denavit-Hartenberg approach to FK.

Angles are in degrees. Lengths are in mm.
"""
import numpy as np


def rot2(theta):
    """3x3 rotation matrix for the flat (2D) arm. Counter-clockwise by theta
degrees."""
    t = np.radians(theta)
    return np.array([[np.cos(t), -np.sin(t), 0],
                     [np.sin(t),  np.cos(t), 0],
                     [0,          0,         1]])
```

```
def move2(dx, dy):
    """3x3 matrix that moves a point by (dx, dy)."""
    return np.array([[1, 0, dx],
                     [0, 1, dy],
                     [0, 0, 1]])


def dh_matrix(theta, d, a, alpha):
    """4x4 Denavit-Hartenberg matrix A = Rz(theta) * Tz(d) * Tx(a) * Rx(alpha).

    theta and alpha are in degrees. d and a are in mm.
    """
    t, al = np.radians(theta), np.radians(alpha)
    ct, st = np.cos(t), np.sin(t)
    ca, sa = np.cos(al), np.sin(al)
    return np.array([
        [ct, -st * ca,  st * sa, a * ct],
        [st,  ct * ca, -ct * sa, a * st],
        [0,        sa,       ca,      d],
        [0,         0,        0,      1],
    ])


def position(T):
    """The (x, y, z) position stored in the last column of a 4x4 matrix T.

    Rounded to 2 decimals (mm). Also turns -0.0 into 0.0 so it prints cleanly.
    """
    return [round(float(v), 2) + 0.0 for v in T[:3, 3]]


if __name__ == "__main__":
    # The Lesson 4 worked example: base 45, shoulder 45, elbow 45.
    theta_b, theta_s, theta_e = 45, 45, 45
    A1 = dh_matrix(theta_b, 0, 0, 90)
    A2 = dh_matrix(90 - theta_s, 0, 250, 0)
    A3 = dh_matrix(-theta_e, 0, 250, 0)
    T = A1 @ A2 @ A3
    print(position(T))   # expected: [301.78, 301.78, 176.78]
```